

Executive function in children and adolescents with unilateral cerebral palsy

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AIM The aim of this study was to compare executive function in children with left- and right-sided unilateral cerebral palsy (CP) and typically developing children.

METHOD There was a cross-sectional cohort of 46 children with unilateral CP (24 right-side, 22 left-side; 25 males, 21 females; mean age 11y 1mo, SD 2y 5mo) and 20 typically developing children (nine males, 11 females; mean age 10y 10mo, SD 2y 4mo). Four cognitive domains of executive function were assessed: attentional control, cognitive flexibility, goal setting, and information processing. Subtests from the Delis–Kaplan Executive Function System, the Test of Everyday Attention for Children, the Rey–Osterrieth Complex figure, and the Wechsler Intelligence Scale for Children – Fourth Edition were utilized. Between-group differences (right unilateral CP, left unilateral CP, and typically developing children) were examined using analyses of covariance.

RESULTS Children with CP performed significantly more poorly than typically developing children on all executive function measures (aggregate executive function: $F(1,63)=31.16$; $p<0.001$; $\eta^2=0.33$). There were no significant differences between children with left and right unilateral CP, except in the case of inhibition/switching total errors, with children with left unilateral CP making fewer errors than children with right unilateral CP ($F(1,39)=4.14$; $p=0.049$; $\eta^2=0.1$).

INTERPRETATION Children and adolescents with unilateral CP experience difficulties across multiple executive function domains compared with typically developing children, irrespective of the side of hemiplegia. This finding supports an early vulnerability model of early brain injury and has implications for intervention for children with CP.

Cerebral palsy (CP) is the leading cause of childhood physical disability worldwide, with financial expenditure in Australia totalling Aus\$ 1.47 billion in 2007.^{1,2} Unilateral CP is the most common type of CP among children born at term and the second most common type in children born preterm, with an incidence of 1 in 1300 live births.^{1,3} Children with unilateral CP experience one-sided motor and postural deficits and many experience difficulties with emotional and behavioural skills⁴ and cognitive functions.⁵ Despite this, there is a paucity of research examining the neuropsychological outcomes of these children.

Executive function is necessary for the successful completion of everyday activities and is an umbrella term that encompasses skills necessary for novel, goal-directed, and complex activity including self-regulation, problem solving, and organization.⁶ Deficits may manifest as an inability to focus and attend to tasks, perseveration, increased errors without subsequent self-correction, and taking longer to complete tasks.⁶ There is no one way to define or concep-

tualize executive function as, by its very nature, it comprises a multitude of higher-order cognitive skills. A conceptual executive function framework in children and adolescents was proposed by Anderson,⁷ and operationalizes executive function as an overall control system comprising four distinct, yet interrelated, components: (1) attentional control, which involves the ability to maintain and focus attention for extended periods; (2) cognitive flexibility, which involves the ability to correct and learn from errors and the flexibly to shift from one response set to another; (3) goal setting, which involves the ability to generate novel goals, plan actions, and complete tasks in a proficient manner; and (4) information processing, which involves the ability to fluently and efficiently complete tasks.

Findings from functional neuroimaging studies have indicated that executive functions are predominantly mediated by the frontal lobes, particularly the prefrontal cortex.⁸ The frontal lobes are the last brain region to reach

maturity, typically by the end of the second decade of life.⁸ The refinement of white matter tracts is important for prefrontal maturation and, in turn, in the progression of executive function.⁸ Recent research has also established that executive dysfunction can be present following early injury to any region of the brain – it does not need to be a frontal lobe lesion – highlighting that the young brain is uniquely vulnerable to early insult in relation to more complex cognitive tasks.⁹ Consequently, the examination of executive function among children with unilateral CP, a condition defined by congenital brain injury,¹ is warranted.

Recent research has started to examine executive function among children with CP.^{10–15} In one study, children aged 9 to 13 years with unilateral ($n=14$) or bilateral CP ($n=18$) were found to have attentional and executive function impairments.¹¹ Specifically, on subtests from the Test of Everyday Attention for Children (TEA-Ch), children with CP performed significantly below age-based norms on subtests assessing sustained and divided attention. Further, on the Behaviour Rating Inventory of Executive Function, Teacher Version, impairments were found in all components, with the greatest impairments on inhibition and shifting.

Similarly, Pirila et al.¹⁰ found deficits in their cohort of children with CP (8–17y; eight with unilateral CP, nine with bilateral CP) on attention, impulsivity, and vigilance outcomes from the Conners' Continuous Performance Test. Working memory deficits among children with unilateral, bilateral, and ataxic CP have also been shown, with lower working memory skills predicting poorer arithmetic skills.^{12,13} The Wisconsin Card Sorting Test has also been used to examine executive function in children with CP (37 unilateral CP, 15 bilateral CP) and it was found that, compared with typically developing children, children with CP had more difficulty completing the task.¹⁴ However, all of the above studies are limited as they examined groups with mixed types of CP and, in most studies, only one of the former components of executive function. Further, more consideration of the developmental context associated with executive function is limited, and potential influences of laterality on executive function in children with left or right unilateral CP have not been considered.

The primary aim of the current study was to determine the pattern of executive function in children and adolescents with unilateral CP. It was hypothesized that, compared with typically developing children, children with unilateral CP would demonstrate poorer executive function across the four domains assessed: attentional control, cognitive flexibility, goal setting, and information processing. The secondary aim was to examine if there was a difference in executive function based on laterality in unilateral CP.

METHOD

Participants

The methods used in this study are reported in the study protocol paper by Bodimeade et al.¹⁶ Institutional ethics

What this paper adds

- This is the first study to examine multiple components of executive function in unilateral CP.
- Children and adolescents with unilateral CP exhibit global executive dysfunction.
- Executive functioning should be assessed as deficits may impede day-to-day functioning and independence.

approval was obtained from the School of Psychology Ethics Committee at the University of Queensland, Brisbane, Australia (10-PSYCH-DCP-32-JM) and from the Queensland Children's Health Services Human Research Ethics Committee, Brisbane, Australia (HREC/10/QRCH/31). Participants were a prospective group of children aged 8 to 16 years, who were divided into two clinical groups: (1) children with unilateral CP (both left and right) and (2) typically developing children. The inclusion period was November 2010 to June 2011. A sample size of 21 in each of the three groups (i.e. left unilateral CP, right unilateral CP, and typically developing children) was calculated to have at least 80% power to detect a large effect size utilizing analysis of covariance (ANCOVA) with three comparison groups.¹⁷

Unilateral CP group

Children were included if they had a diagnosis of unilateral CP, could follow instructions, and had English as their first language. Children were ineligible if they had an uncontrolled seizure disorder.

Comparison group

Typically developing children (matched for age and sex) were recruited as a reference sample. Children were either typically developing siblings or friends of children from the unilateral CP group. Children were ineligible if English was not their first language or if they had a history of developmental, neurological, physical, or psychiatric conditions. Recruitment also occurred through staff newsletters and from other studies within the centre. A provisional psychologist conducted a brief telephone screening interview to ensure they met the study selection criteria.

Design and procedure

A cross-sectional design with three groups was utilized: (1) children with right unilateral CP; (2) children with left unilateral CP; and (3) typically developing children who were matched for age and sex. Outcome measures were a battery of executive function measures assessing cognitive flexibility, goal setting, attentional control, and information processing.

Background measures

All parents completed a demographic questionnaire, which collected basic demographic information (e.g. contact details), as well as medical and developmental information (e.g. comorbid diagnoses). Children's motor functioning was classified using the Gross Motor Function Classification System (GMFCS)¹⁸ and the Manual Ability Classifica-

tion System (MACS).¹⁹ Intellectual function was measured using the Wechsler Abbreviated Scale of Intelligence²⁰ as a sample descriptor (mean 100, SD 15).

Executive function measures

The Rey–Osterrieth Complex Figure Test²¹ and subtests from the Delis–Kaplan Executive Function System,²² the TEA-Ch,²³ and the Wechsler Intelligence Scale for Children–Fourth Edition²⁴ were utilized to assess executive function components (attentional control, cognitive flexibility, goal setting, and information processing; see Fig. 1). All children completed the assessments using their unaffected hand. A full description of each measure for this study has previously been published.¹⁶

Statistical analyses

Between-group differences were examined in a series of analyses of covariance (ANCOVAs). Preliminary analyses used Fisher's exact test. All statistical analyses were performed using SPSS version 18 (IBM SPSS Statistics,

IBM Corp. NY, USA) and R version 2.9.1. (R Foundation for Statistical Computing, www.r-project.org)

RESULTS

A total of 188 children and adolescents from the Queensland Cerebral Palsy & Rehabilitation Research Centre database diagnosed with unilateral CP were assessed for eligibility. Forty-six children and adolescents with unilateral CP (24 right-side, 22 left-side; 25 males, 21 females; mean age 11y 1mo, SD 2y 5mo) were selected to take part along with 20 typically developing children (nine males, 11 females; mean age 10y 10mo, SD 2y 4mo; Fig. 2).

Demographic information

Overall, children with unilateral CP were more likely to have some form of learning disorder (13 out of 46), compared with none of the typically developing children ($p=0.014$). Children with unilateral CP were also more likely to have visual difficulties (15 out of 46) than typically developing children (0 out of 20; $p=0.04$) and were more

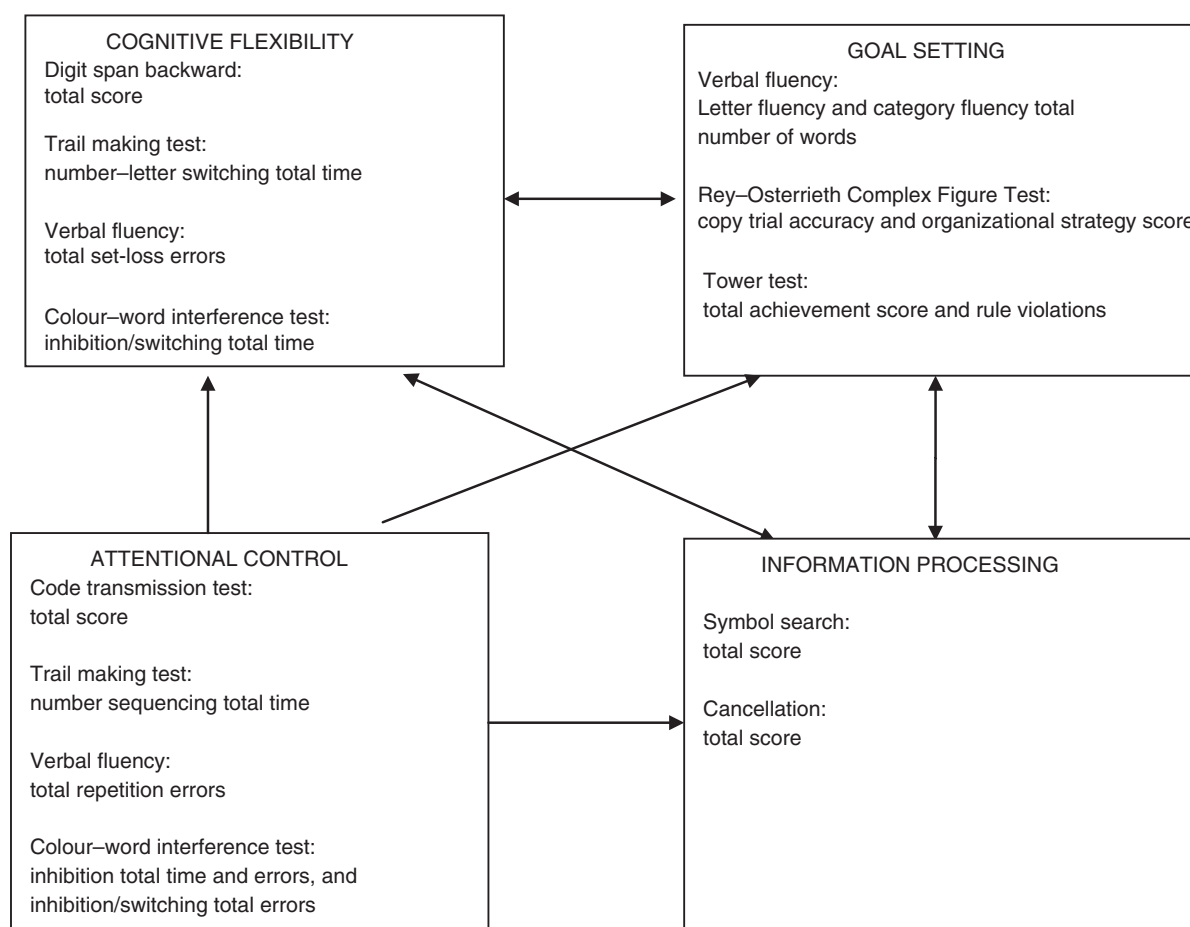


Figure 1: Model of executive function with corresponding neuropsychological assessments. Digit span backward from the Wechsler Intelligence Scale for Children – Fourth Edition (WISC-IV); trail making test, verbal fluency, colour-word interference test, and tower test from the Delis–Kaplan Executive Function System; code transmission test from the Test of Everyday Attention for Children; symbol search and cancellation from the WISC-IV.

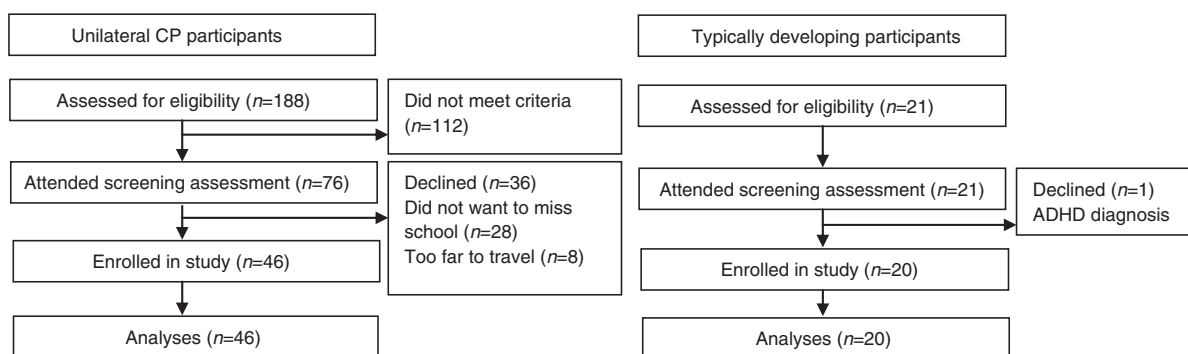


Figure 2: Study flow chart. ADHD, attention-deficit-hyperactivity disorder.

likely than typically developing children to be living in single-parent families or with a step-parent ($p=0.020$; Table I). The children with unilateral CP were predominantly classified in GMFCS level I, walking independently without a gait aid, and not requiring a handrail with stairs. The participants were also predominantly classified as MACS level II and hence able to handle most objects but with somewhat reduced quality and/or speed of achievement so that alternative ways of performance may be used.

Preliminary analyses

Small proportions of randomly missing data due to colour blindness and non-compliance were managed with pairwise analyses. Analysis with outliers removed and cases Winsorized at the 95th centile did not change interpretation and thus these cases were retained for the main analysis. Several measures showed significant skew within groups. Again, results did not differ as a result of the transformation; therefore, untransformed data are presented. In keeping with current discussions in the literature,^{25,26} it was decided not to control for IQ, as a strong relationship between IQ and executive function is common in the context of early brain injury. In our study, the correlation between IQ and the aggregate executive function composite was in keeping with previous findings ($r=0.7$).

Main analyses

A series of ANCOVAs were conducted controlling for age. Controlling for gross family income did not substantively change analysis, and thus only age was controlled in the reported analysis. Significant between-group differences were followed up with a priori linear contrasts, comparing the typically developing children with all children with unilateral CP and comparing participants with left and right unilateral CP. The means and standard deviations of each group, as well as the overall ANCOVAs, are reported in Table II and follow-up linear contrasts are reported in Table III.

Attentional control

There were significant between-group differences for all attentional control measures with the exception of verbal

fluency total repetition errors and inhibition total errors, and there was a marginally significant between-group difference ($p=0.064$) in inhibition total errors. Children with unilateral CP performed significantly worse than typically developing children on code transmission total correct, inhibition total time and total errors, inhibition/switching total errors, and number sequencing total time. There was a difference between children with left and right unilateral CP on inhibition/switching total errors, such that children with left unilateral CP produced significantly fewer errors, but this difference was statistically significant only at an alpha level of 0.05 that fails to adequately control for the family-wise error.

Cognitive flexibility

There were significant between-group differences on digit span backwards and number-letter switching total time. There was also a marginally significant ($p=0.067$) between-group difference on inhibition/switching total time, but no significant difference in verbal fluency total set-loss errors. Children with unilateral CP performed significantly worse than typically developing children on digit span backwards, number-letter switching total time, and inhibition/switching total time. There were no significant differences between children with left and right unilateral CP on any cognitive flexibility measure.

Goal setting

Analyses revealed significant between-group differences for all goal-setting measures. Children with unilateral CP performed significantly worse than typically developing children on letter and category fluency total correct, Rey figure copy trial accuracy and organizational strategy score, and tower test achievement and rule violations. There were no differences between children with left and right unilateral CP on any goal-setting measure.

Information processing

There were significant between-group differences on all information processing measures. Children with unilateral CP performed significantly worse than typically developing children on both symbol search and cancellation. Again,

Table 1: Characteristics of participants

Characteristics	Right unilateral CP (<i>n</i> =22)	Left unilateral CP (<i>n</i> =24)	Typically developing (<i>n</i> =20)
Mean age, <i>y</i> (SD)	11.09 (2.54)	11.08 (2.38)	10.8 (2.29)
Sex, <i>n</i>			
Male	12	13	9
Female	10	11	11
GMFCS, <i>n</i>			
Level I	14	21	n/a
Level II	8	3	n/a
MACS, <i>n</i>			
Level I	1	5	n/a
Level II	21	19	n/a
Handedness, <i>n</i>			
Right	0	24	17
Left	21	0	3
Ambidextrous	1	0	0
Mean intellectual functioning (SD)			
Full-scale IQ	84.95 (14.65)	86.75 (17.96)	116.5 (10.7)
Verbal IQ	87.23 (16.14)	91.12 (20.83)	118.25 (15.41)
Performance IQ	85.18 (13.45)	84.96 (15.03)	111.25 (11.29)
Ethnicity, <i>n</i>			
Caucasian Australian	21	22	20
New Zealand Maori	0	1	0
Lebanese Australian	0	1	0
Chinese Australian	1	0	0 (0)
Other diagnoses, <i>n</i>			
Seizure disorder (controlled)	3	4	0
Learning disorder	6	7	0
Hearing impairment (corrected)	1	3	0
Vision impairment (corrected)	6	9	0
Attention-deficit-hyperactivity disorder	3	3	0
Autistic spectrum disorder	2	3	0
Anxiety disorder	1	0	0 (0)
Perthes disease	1	0	0
Medication use, <i>n</i>			
Yes	7	4	0
No	15	20	20
Gross household income (Aus\$), <i>n</i>			
Under 5,000	1	2	0
25,000–50,000	3	3	2
50,000–75,000	4	4	2
75,000–100,000	5	8	3
Over 100,000	8	6	11
Family type child living in, <i>n</i>			
Original	13	15	18
Step-family	7	2	1
Sole parent	1	6	1
Joint custody	0	0 (0)	0
Parental marital status, <i>n</i>			
Married	17	17	19
Single	1	4	0
De facto ^a	4	0	0
Divorced or separated	0 (0)	3 (13)	1
Parental education, <i>n</i>			
Some high school	6	5	3
Completed high school	6	7	3
Post high school training	4	6	1
University degree	5	5	10
Postgraduate degree	1	1	5

^aUnmarried, unrelated adults living together. GMFCS, Gross Motor Function Classification System; MACS, manual ability classification system; n/a, not applicable.

there were no significant differences between children with right and left unilateral CP on either task.

Aggregate executive function composite

All executive function variables were standardized and averaged to form an aggregate executive function composite score with high internal consistency ($\alpha=0.93$). There was a significant between-group difference on this score,

with unilateral CP participants performing significantly worse than typically developing children. There was no significant difference between children with left and right unilateral CP on the total executive function aggregate.

DISCUSSION

Children with unilateral CP performed significantly worse than typically developing children on the executive func-

Table II: Means (SD) of each group for neuropsychological measures

Domain	Measure	Typically developing children (n=20)	Left unilateral CP (n=24)	Right unilateral CP (n=22)	F	p	η^2
Cognitive flexibility	Digit span backward total scaled score	11.1 (2.25)	8.12 (2.97)	7.82 (2.63)	9.75	<0.001***	0.24
	Number-letter switching total time (s)	97.05 (39.19)	186 (76.63)	188.41 (125.95)	7.96	0.001***	0.21
	Verbal fluency set-loss errors raw score	1.74 (1.79)	3.38 (4.05)	3.64 (3.06)	2.34	0.105	0.07
Goal setting	Inhibition/switching total time (s)	82.85 (22.95)	102.3 (38.32)	105.26 (36.39)	2.83	0.067 [†]	0.09
	Letter fluency number correct raw score	26.5 (7.86)	18.42 (10.37)	13.55 (8.24)	13.46	<0.001***	0.3
	Category fluency number correct raw score	32 (7.71)	25.29 (10.02)	24 (7.36)	7.6	0.001***	0.2
	Rey copy trial accuracy score	35.05 (1.63)	25.02 (11.36)	25.83 (11.87)	8.05	0.001***	0.21
	Rey organizational strategy score	5.2 (1.06)	3.83 (1.71)	3.43 (1.36)	11.44	<0.001***	0.27
	Tower test achievement raw score	16.25 (3.78)	12.57 (3.63)	12.59 (4.04)	6.57	0.003**	0.18
Attentional control	Tower test rule violations raw score	0.9 (1.55)	8.43 (10.23)	6.45 (4.81)	7.76	0.001***	0.2
	Code transmission total correct raw score	36.55 (3.63)	27.08 (12.7)	28.86 (11.57)	5.74	0.005**	0.16
	Verbal fluency repetition errors raw score	1.47 (1.61)	0.92 (0.97)	1.18 (1.05)	1.14	0.327	0.04
	Inhibition total time (s)	82.7 (30.5)	113.78 (39.75)	110.84 (39.78)	5.46	0.007**	0.16
	Inhibition total errors raw score	4.45 (4.07)	7.13 (6.72)	8.47 (6.27)	2.88	0.064 ^a	0.09
	Inhibition/switching total errors raw score	3.35 (2.81)	4.83 (5.37)	7.95 (5.44)	5.71	0.005**	0.16
Information processing	Number sequencing total time (s)	38.7 (13.45)	69.52 (29.57)	62.09 (32.07)	8.27	0.001***	0.21
	Symbol search raw score	25.65 (4.88)	17.04 (7.32)	18.59 (6.37)	12.9	<0.001***	0.29
	Cancellation raw score	72.95 (21.51)	60.43 (24.21)	59.86 (14.9)	3.44	0.039*	0.1
	Total execute function aggregate	0.45 (0.34)	-0.21 (0.68)	-0.24 (0.52)	15.39	<0.001***	0.33

Digit span backward, symbol search, and cancellation from the Wechsler Intelligence Scale for Children – Fourth Edition. Number sequencing and number-letter switching from trail making test from the Delis-Kaplan Executive Function System (D-KEFS). Verbal fluency from the D-KEFS. Inhibition and inhibition/switching from the colour-word interference test from the D-KEFS. Letter fluency and category fluency from verbal fluency from the D-KEFS. Code transmission test from the Test of Everyday Attention for Children. ^a $p < 0.07$. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. CP, cerebral palsy; Rey, Rey-Osterrieth Complex Figure Test.

tion composite, as well as across all four domains of executive function – attentional control, cognitive flexibility, goal setting, and information processing – consistent with hypotheses and previous research.^{11,13} Particularly strong effect sizes were found on letter fluency and the Rey Figure organizational strategy score, suggesting that children with unilateral CP may have particular difficulty with the goal-setting domain of executive function.

Inconsistent with the hypotheses, children with unilateral CP demonstrated preserved verbal self-monitoring. This may be the result of the conservation of language and verbal skills. Previous research has shown preservation of verbal over other cognitive functions following early brain injury.²⁷ Furthermore, sparing of verbal self-monitoring skills may also be reflective of a functional reorganization of verbal skills, leading to ‘cognitive crowding’: the conservation of verbal skills among children with both early left- and early right-hemisphere damage to the detriment of non-verbal skills.²⁸ Consistent with this explanation is the fact that children with unilateral CP in our study performed significantly worse than typically developing children on non-verbal self-monitoring tasks. However, a definite explanation of these findings from this study alone is not possible, and these results may simply be reflective of continuing maturation within the typically developing children.

This study did not find evidence for the laterality of executive function according to side of unilateral CP.

Across all domains of executive function, performance for children with left and right unilateral CP was comparable. The only exception was the inhibition/switching task within the attentional control domain, on which children with right unilateral CP committed significantly more errors. However, this finding was significant at an alpha level of 0.05, and hence may be spurious. Future research, with a larger sample size, is needed to confirm if this is a genuine laterality effect. If it is found to be genuine, then it may be reflective of a higher rate of reading impairments in children with right unilateral CP²⁹ as the inhibition/switching task was the only measure that included reading skills.

Clinical practice continues to focus on the physical difficulties faced by children with unilateral CP.³⁰ However, this study demonstrates that children with mild unilateral CP show significant executive function impairments in comparison to typically developing children. Impairments in executive function, to the degree shown in this study, are clinically significant and could lead to functional difficulties in any goal-directed, novel, or complex activity including academic achievement. It is important that neuropsychological research examines specific cognitive skills, such as executive function, rather than just global cognitive measures such as IQ. Mapping the executive function profile of children with unilateral CP provides important information that can be utilized in rehabilitation. It is imperative that ongoing neuropsychological

Table III: Linear contrasts for each neuropsychological measure

Domain	Measure	Linear contrast	Mean difference	F	p	η^2
Cognitive flexibility	Digit span backward total scaled score	TDC vs combined unilateral CP	3.13	19.61	<0.001 ***	0.24
		Left unilateral CP vs right unilateral CP	0.30	0.14	0.714	<0.001
	Number-letter switching total time (s)	TDC vs combined unilateral CP	-90.1	16.17	<0.001 ***	0.21
		Left unilateral CP vs right unilateral CP	-2.41	0.01	0.935	<0.001
	Inhibition/switching total time (s)	TDC vs combined unilateral CP	-20.87	5.66	0.021 *	0.09
Goal setting		Left unilateral CP vs right unilateral CP	-2.96	0.07	0.796	<0.001
	Letter fluency number correct raw score	TDC vs combined unilateral CP	10.41	21.73	<0.001 ***	0.26
		Left unilateral CP vs right unilateral CP	4.87	3.47	0.069	0.07
	Category fluency number correct raw score	TDC vs combined unilateral CP	7.33	14.96	<0.001 ***	0.19
		Left unilateral CP vs right unilateral CP	1.29	0.32	0.572	0.01
	Rey copy trial accuracy score	TDC vs combined unilateral CP	9.64	16.23	<0.001 ***	0.21
		Left unilateral CP vs right unilateral CP	-0.81	0.07	0.795	<0.001
	Rey organizational strategy score	TDC vs combined unilateral CP	1.56	21.70	<0.001 ***	0.02
		Left unilateral CP vs right unilateral CP	0.40	0.98	0.327	0.18
	Tower test achievement raw score	TDC vs combined unilateral CP	3.67	13.36	<0.001 ***	<0.001
		Left unilateral CP vs right unilateral CP	-0.02	<0.001	0.982	0.19
	Tower test rule violations raw score	TDC vs combined unilateral CP	-6.56	14.43	<0.001 ***	0.02
		Left unilateral CP vs right unilateral CP	1.98	0.77	0.386	0.15
	Code transmission total correct raw score	TDC vs combined unilateral CP	8.62	11.19	<0.001 ***	0.01
		Left unilateral CP vs right unilateral CP	-1.78	0.28	0.599	0.16
Attentional control	Inhibition total time (s)	TDC vs combined unilateral CP	-29.67	11.00	0.002 **	0.08
		Left unilateral CP vs right unilateral CP	2.94	0.06	0.801	<0.001
	Inhibition total errors raw score	TDC vs combined unilateral CP	-3.32	5.15	0.027 **	0.01
		Left unilateral CP vs right unilateral CP	-1.34	0.52	0.475	0.09
	Inhibition/switching total errors raw score	TDC vs combined unilateral CP	-2.96	5.63	0.021 **	0.10
		Left unilateral CP vs right unilateral CP	-3.12	4.14	0.049 *	0.20
	Number sequencing total time (s)	TDC vs combined unilateral CP	-27.27	15.60	<0.001 ***	0.02
		Left unilateral CP vs right unilateral CP	7.43	0.69	0.411	0.28
	Symbol search raw score	TDC vs combined unilateral CP	7.87	25.08	<0.001 ***	0.01
		Left unilateral CP vs right unilateral CP	-1.55	0.64	0.430	0.10
Information processing	Cancellation raw score	TDC vs combined unilateral CP	12.79	6.98	0.010 **	<0.001
		Left unilateral CP vs right unilateral CP	0.57	0.01	0.916	0.33
	Executive function composite score	TDC vs combined unilateral CP	0.67	31.16	<0.001 ***	<0.001
Executive function aggregate		Left unilateral CP vs right unilateral CP	0.03	0.05	0.819	<0.001

Measures that were not significant on the ANCOVA were not followed up with linear contrasts and hence are not included in this table (see Table II). Digit span backward, symbol search, and cancellation from the Wechsler Intelligence Scale for Children – Fourth Edition. Number sequencing and number-letter switching from trail making test from the Delis-Kaplan Executive Function System (D-KEFS). Inhibition and inhibition/switching from the colour-word interference test from D-KEFS. Letter fluency and category fluency from verbal fluency from the D-KEFS. Code transmission test from the Test of Everyday Attention for Children. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$. TDC, typically developing children; CP, cerebral palsy; Rey, Rey-Osterrieth Complex Figure Test.

surveillance and early rehabilitation for executive function become part of standard care for children with CP. Additional research is needed to identify effective early rehabilitation interventions and to explore the relationship between executive function and lesion site to make earlier, specialized executive function rehabilitation possible in the future.

To our knowledge, this is the first study to examine executive function in children with CP by focusing on a specific CP group (unilateral CP) and using a paediatric multifaceted model of executive function, as well as the first study to explore laterality of executive function in children with CP. This study is limited by sample size and by the methods used to recruit the typically developing children. It is possible that laterality effects may have been found with a larger sample size and that potential differences between children based on age may have been uncovered. In addition, including siblings in the comparison group of typically developing children may have led to non-independence in statistical analyses. Examination of

lower-order cognitive skills in addition to executive function is an avenue for future research, as is examining the relationship between executive function impairments and functional difficulties in everyday life in this population.

CONCLUSION

Children with unilateral CP show deficits across all executive function domains (i.e. attentional control, cognitive flexibility, goal setting, and information processing), highlighting the detrimental effect of an early unilateral brain injury. There was no evidence of laterality of executive function. Further research on executive function in children with CP is needed so that effective, specialized executive function rehabilitation may become a clinical reality.

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